



REVIEW ARTICLE

Nanoemulgel : An Innovative Drug Delivery System for Improved Therapeutic Effectiveness of Lipophilic Medications

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Received: 21 Aug 2025

Revised: 26 Sep 2025

Accepted: 19 Oct 2025

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ABSTRACT

Lipophilic drugs are often constrained by poor aqueous solubility, variable absorption, and limited oral bioavailability, which hinder their clinical effectiveness. To overcome these challenges, novel delivery systems are being developed. This review explores the nanoemulgel delivery system as a promising platform aimed at boosting solubility, stability, and therapeutic performance of lipophilic drugs. Nanoemulgels are formulated by incorporating nanoemulsions composed of lipophilic drugs into a hydrogel base, resulting in a semi-solid matrix that facilitates enhanced transdermal and mucosal drug delivery. This approach merges the advantages of nanoemulsions and gels, offering benefits like sustained drug release and enhanced therapeutic performance. Nanoemulgels have demonstrated potential in the treatment of dermatological conditions [e.g., acne, psoriasis, fungal infections], inflammatory diseases [e.g., osteoarthritis, rheumatoid arthritis], and have shown applicability in alternative routes of administration including ocular, vaginal, dental, and intranasal delivery. These routes expand the therapeutic scope to systemic and localized Conditions including hair loss [alopecia], gum disease [periodontitis], and Parkinson's disease. Nanoemulgel-based drug delivery represents a significant advancement in pharmaceutical sciences, particularly for lipophilic compounds. Its ability to improve drug bioavailability and target delivery opens new avenues in both topical and systemic therapies..

Keywords: Nanoemulgel, Transdermal Drug delivery system, Transdermal therapy, Nanoemulsion, Hydrogels, Controlled drug release.





INTRODUCTION

Drug delivery systems [DDS] aim to improve therapeutic outcomes by controlling drug release and targeting specific sites. They include oral, transdermal, intravenous, mucosal, and inhalation methods. Transdermal drug delivery systems [TDDS] are non-invasive formulations, often designed as patches, that administer medication through the skin and deliver drugs into circulation. The first approved TDDS, Transderm-Scop [1980], used scopolamine for motion sickness via a membrane-controlled system. Despite the skin barrier, especially the stratum corneum, advancements in formulation have enhanced permeability, making TDDS a convenient and effective option. Key examples are listed in Table 1 [1].

Positive and negative aspects of TDDS

Transdermal drug delivery offers several benefits, including the avoidance of first-pass metabolism and gastrointestinal degradation, leading to improved bioavailability. It enhances therapeutic effectiveness, reduces dosing frequency, and improves patient adherence. However, it may cause skin irritation in some users and is generally suitable only for potent, non-ionized drugs with a molecular weight below 500 Daltons due to the skin's selective permeability.

Anatomy and physiology of skin

The skin comprises three main layers—epidermis, dermis, and hypodermis—each with specialized functions. The epidermis serves as the body's first line of defense against microbes, UV radiation, and water loss. It primarily consists of keratinocytes and melanocytes, arranged in layers: the stratum basale [mitotically active cells], spinosum [providing strength via desmosomes], granulosum [rich in keratohyalin and lipids], lucidum [in thick skin], and corneum, which is made of dead corneocytes and provides barrier and hydration control. The stratum corneum ranges from 10–15 μm when dry to 40 μm when hydrated [2–4]. The dermis, located beneath the epidermis, is a connective tissue layer that supports skin strength, elasticity, and sensory function. It is divided into the papillary layer, containing loose connective tissue and dermal papillae, and the reticular layer, rich in collagen, elastin, and specialized elastic fibers like elastin and oxytalan [5]. The hypodermis, or subcutaneous layer, is mainly composed of adipose and connective tissues. It functions in insulation, energy storage, and mechanical cushioning, and it houses fibroblasts, blood and lymph vessels, nerves, sweat glands, and hair follicles [6].

Basic components of TDDS

1. The polymer matrix in TDDS regulates drug release and must be biocompatible, stable, and suitable for embedding drugs with favorable diffusion properties [7–9].
2. Penetration enhancers increase skin permeability by altering lipid structure and should be safe, inert, and compatible [10].
3. Pressure-sensitive adhesives [PSAs] ensure patch adhesion with minimal pressure and must be non-irritant and residue-free [11].
4. The backing layer provides flexibility, chemical resistance, and moisture control, enhancing patch durability and comfort [12].
5. The release liner protects the patch pre-application, made from occlusive or non-occlusive materials with release coatings and solvents/plasticizers for drug dispersion and flexibility [13].

TDDS can be classified into three types based on their design and structure:

1. Reservoir system: In this system, a drug reservoir is positioned between the backing layer and a rate-controlling membrane. The drug is typically suspended, gelled, or dissolved within a polymer matrix.[14]
2. Matrix system: The drug is dispersed throughout the polymer matrix, and the release rate is controlled by diffusion through the matrix material.[15]
3. Micro Reservoir system: This system features small reservoirs of the drug suspended within a polymer matrix, providing controlled release via diffusion.[16]

In a matrix system, the drug is embedded within a semisolid matrix as a solution or suspension. This drug-containing layer lies in direct contact with the release liner, and an adhesive layer supports the overall structure



**Kousalya et al.,**

partially overlays the drug layer.^[17] Micro Reservoir system ^[18]The Micro Reservoir system, as described by Bathe and Kapoor [2015], integrates matrix dispersion with a reservoir mechanism. Initially, the drug is solubilized in an aqueous solution containing a water-soluble polymer, after which it is dispersed within lipophilic polymers to create microscopic spheres that serve as drug reservoirs.^[19]

Evaluation of Transdermal Films:

- A. Drug Content Uniformity: A defined number of patches are dissolved, filtered, and analyzed by UV spectroscopy or HPLC to determine consistent drug distribution [20].
- B. Surface pH: The surface pH of patches is measured to ensure compatibility with skin and minimize irritation potential [21].
- C. Folding Endurance: The patch is repeatedly folded at the same spot until it breaks. The total number of folds before breaking reflects its mechanical durability [22].
- D. Patch Thickness: Measured using a micrometer, with weight recorded to assess uniformity and consistency [23].
- E. Moisture Content: Patches are dried in a desiccator, and the weight loss is used to calculate residual moisture percentage [24].
- F. Water Absorption: After drying, patches are exposed to humidity and weight gain is measured to determine moisture uptake capacity [25].
- G. Drug-Carrier Interaction: Interactions between drug and excipients are assessed using analytical techniques like TLC or HPLC [26].
- H. Tack Property: The adhesive ability of the patch under minimal pressure is evaluated, influenced by polymer type and molecular weight [27].
- I. Tensile Strength: The force required to break the patch is measured by suspending it under tension using a pulley system [28].
- J. In Vitro Drug Release: Conducted using Franz diffusion cells, where the release of drug from the patch into a receptor medium is monitored under controlled temperature and stirring [29].
- K. In Vitro Permeation: Excised Wistar rat abdominal skin is mounted on diffusion cells to study drug permeation. Samples are periodically collected and analyzed [30].
- L. Skin Irritation Test: Patches are applied to rabbit skin, and signs of irritation are observed after 24 hours to assess dermal safety [31].
- M. Stability Studies: Conducted under ICH-recommended conditions [$40^{\circ}\text{C} \pm 0.5^{\circ}\text{C}$, 75% RH] for 6 months. Drug content is evaluated at specific time points [0, 30, 60, 90, and 180 days] to ensure stability [32].

NANOEMULSION

Nanotechnology, operating at scales between 0.1 and 100 nm, has significantly influenced the pharmaceutical field, leading to the development of nanopharmaceuticals—drug formulations utilizing nanoscale technologies [Shah & Bhalodia, 2011] [33]. These include systems such as nanoemulsions, nanospheres, nanosuspensions, nanocapsules, lipid nanoparticles, dendrimers, and more. Among these, nanoemulsions have emerged as a promising drug delivery platform due to their enhanced kinetic stability, resistance to temperature changes and dilution, and nanometer-sized droplets [5–200 nm] [34]. These are composed of two immiscible phases—typically oil and water—stabilized by surfactants and co-surfactants.

Nanoemulsions offer optical clarity, long-term stability, and versatility in administration, being suitable for oral, ocular, and transdermal delivery. They are extensively utilized for delivering vaccines, gene therapies, and cosmeceutical agents, making them valuable across both therapeutic and cosmetic sectors [35]. A comprehensive review of recent research on nanoemulsions, their composition, methods of preparation, and therapeutic outcomes has been summarized in Table 3.



**Kousalya et al.,****Types of nanoemulsion**

Based on Components of Nano emulsions

Oil-in-water [o/w] nanoemulsions consist of oil droplets that are uniformly distributed within a continuous aqueous phase. Conversely, water-in-oil [w/o] nanoemulsions feature water droplets that are dispersed throughout a continuous oil phase. Additionally, bi-continuous nanoemulsions are characterized by the presence of microdomains of both oil and water intermingled within the system.[36]

Components Of Nanoemulsion

Main three components of Nano-emulsions are

- Oil
- Surfactant/ co surfactants
- Aqueous phase.

Method of preparation of nano emulsion:**High-Energy Methods**

These involve mechanical force to reduce droplet size:

- High-Pressure Homogenization: Forces the emulsion through a narrow gap under high pressure [500–5000 psi], producing nanosized droplets through shear and cavitation.
- Ultrasonication: Uses high-frequency sound waves to break droplets into nano-size via cavitation.
- High-Speed Homogenization: Employs a rotor-stator system to create shear forces, often used in initial emulsification.[37]

Low-Energy Methods

These rely on internal system properties, needing minimal mechanical input:

- Spontaneous Emulsification: Involves mixing an organic phase [oil + surfactant] into an aqueous phase, forming nano-droplets via interfacial turbulence.
- Phase Inversion Composition [PIC]: Water is gradually added to oil-surfactant mix, causing transition from w/o to o/w nanoemulsion.
- Phase Inversion Temperature [PIT]: Temperature is adjusted to alter surfactant behavior, promoting phase inversion and nanoemulsion formation.[38]

Characterization of Nanoemulsions

Several factors influence the characterization of nanoemulsions, such as particle size, viscosity, density, phase inversion, turbidity, refractive index, and skin permeation. Techniques used to assess these characteristics include those outlined by Shah and Bhalodia [2011].[39]

Thermal Conductivity Technique

One method for measuring thermal conductivity in nanoemulsions is the 3-wire technique, where a thermal conduct meter monitors the temperature of a metal wire over time. Water-based [O/E] nanoemulsions exhibit high thermal conductivity, as noted by Devarajan and Ravichandran [2011].[40]

Dynamic Light Scattering [DLS]

Dynamic light scattering is performed using a neon laser with a wavelength of 632 nm at a 90° angle to determine the particle size and distribution of the nanoemulsions, as described by Wang et al. [2004].[41]

Zeta Potential

Zeta potential is measured to assess the surface charge of nanoemulsions using a mini electrode, as outlined by Bhatt, P. and S. Madhav [2011].[42]



**Kousalya et al.,****Transmission Electron Microscopy [TEM]**

Transmission Electron Microscopy [TEM] is used to determine the dimensions, quantity, mass, and morphology of nanoemulsions. For oil-in-water [O/W] nanoemulsions, samples are treated with uranyl acetate and placed on a polymer-coated grid. The samples are then examined after water evaporation.[43]

Drug Content

Drug content in nanoemulsions is quantified using the Western Blot technique. The viscosity of nanoemulsions is evaluated using a rotary viscometer, as these emulsions typically exhibit low viscosity under different conditions of temperature and shear rates.[44]

Phase Analysis Technique

Phase analysis is crucial for nanoemulsions, which are often formulated using phase inversion or self-emulsification methods, to assess their dispersibility and stability.[45]

Advantages and Disadvantages of Nanoemulsions

Advantages: Nanoemulsions enhance drug absorption and bioavailability, support multiple delivery routes [oral, topical, IV], and enable faster penetration. They are also useful for taste masking, expanding their applicability [46]. **Disadvantages:** Challenges include limited solubility for high-melting-point drugs, the need for safe surfactants, and instability under environmental changes like temperature and pH shifts [Figuerola Alvarez & Blanco Méndez, 2001]. [47]

NANOEMULGEL

Nanoemulgel is a cutting-edge drug delivery system that combines a nanoemulsion—an oil-in-water or water-in-oil emulsion with nanometer-sized droplets—with a gel base. This hybrid formulation enhances the stability, skin penetration, and bioavailability of active pharmaceutical ingredients [APIs], making it particularly useful for topical and transdermal applications. Nanoemulgels are ideal for delivering drugs that are poorly soluble in water, providing benefits such as improved drug release, targeted delivery, and greater patient compliance due to their non-greasy, easily spreadable texture. [Aithal G.C., Narayan R., Nayak U.Y. 2021; Pund S., et al. 2015]^[48]

Advantages of nanoemulgels

Effective Incorporation of Lipophilic Drugs: Nanoemulgels enhance the solubility of lipophilic [fat-soluble] drugs, which are typically challenging to deliver through traditional systems.

Improved Loading Capacity: The combination of nanoemulsion and gel allows for a higher loading capacity, making it possible to deliver greater amounts of active ingredients.

Better Patient Compliance: Their smooth, non-greasy texture makes them more comfortable for users, leading to increased patient adherence to treatment protocols.

Enhanced Drug Permeability: Nanoemulgels enhance the ability of drugs to penetrate the skin more effectively, facilitating transdermal drug delivery.^[49]

Application**Pharmaceuticals**

Drug Delivery: Nanoemulgels enhance the bioavailability and controlled release of poorly water-soluble drugs, ensuring targeted delivery to specific organs or tissues. This enhances therapeutic efficacy while reducing the risk of side effects.

Topical Medications

These gels are particularly useful in topical formulations, improving the penetration of active ingredients through the skin, making them ideal for treating dermatological conditions and facilitating transdermal drug delivery [50].



**Kousalya et al.,****Cosmetics**

Skincare Products: Nanoemulgels are employed in cosmetic formulations to more effectively deliver active ingredients such as vitamins, antioxidants, and anti-aging agents. They improve the stability, skin penetration, and release of these compounds, benefiting skincare routines.

Hair Care: In hair care products, nanoemulgels are used to deliver conditioning agents, vitamins, and growth factors to both the hair and scalp, enhancing their overall effectiveness.^[51]

Food Industry:

Nutraceuticals: Nanoemulgels can be utilized to increase the solubility and bioavailability of nutrients and active compounds in food and dietary supplements, improving their stability and release profile, thus enhancing their effectiveness.

Agriculture

Pesticide Delivery: In agriculture, nanoemulgels can improve the delivery of pesticides to plants, enhancing their penetration and effectiveness, while potentially reducing the need for excessive pesticide application.^[52]

Biomedical Applications

Diagnostic Imaging: Nanoemulgels can be used as contrast agents in imaging techniques, such as MRI and ultrasound, improving the visualization and targeting of specific tissues or abnormalities.

Cancer Therapy: These systems are also explored in cancer therapy, where they can be used to deliver chemotherapeutic agents or gene therapies directly to cancer cells, improving therapeutic effects while minimizing damage to healthy tissues.^[53]

Environmental Applications

Pollutant Removal: Nanoemulgels are being considered for use in environmental cleanups, where they can encapsulate and remove pollutants or toxins from water and soil, offering an efficient method for decontamination.^[54]

Formulation components: The formulation of nano-emulgels consists of two primary components: the gelling agent and the nano-emulsion, which is characterized by the presence of nano-sized droplets in either oil-in-water [o/w] or water-in-oil [w/o] configurations. Each emulsion type contains both an aqueous phase and an oily phase.^[55] figure 1 explains the components

Oils

Oil selection is critical in nanoemulgel formulation, influencing solubility, stability, permeability, and viscosity. Traditional edible oils are generally avoided due to poor emulsification and drug solubility [Tiwary et al., 2013]. Labrafac has been effectively used for butenafine nanoemulgels [Rizwan et al., 2009], while Capryol 90 has shown improved stability in nanoemulsions of leflunomide and paclitaxel [Arora et al., 2014]. [56]

Surfactants

Surfactants reduce interfacial tension between oil and water phases, improving emulsion stability. Their selection is based on safety, emulsification efficiency, and compatibility with oils. For example, Tween 20 is compatible with oils like Capryol 90 and oleic acid [Wang et al., 2015]. Non-ionic surfactants are preferred due to their lower irritation potential and better systemic safety profile [Aggarwal et al., 2014]. [57]

Co-Surfactants

Co-surfactants assist surfactants by further lowering interfacial tension and improving emulsification. They enhance interfacial film flexibility and support the formation of stable emulsions [Dev et al., 2015]. Their interaction with surfactants and drug partitioning affects release rates. Common co-surfactants include PEG-400 and Transcutol® HP. [58]



**Kousalya et al.,****Aqueous Solvents and Gelling Agents**

Water and ethanol are commonly used as the aqueous phase in nanoemulsion systems. Gelling agents like Carbopol 934, Carbopol 940, and HPMC are added to increase viscosity, converting the emulsion to a gel. These agents also enhance spreadability, skin retention, and may interact with surfactants to fine-tune formulation properties [Peng et al., 2015]. [59]

Miscellaneous Components

Preservatives such as methylparaben, propylparaben, benzoic acid, and benzalkonium chloride are included to prevent microbial growth. Antioxidants like butylated hydroxytoluene [BHT] are added to enhance formulation stability. [60]

Method of preparation of nanoemulgel

Nanoemulgel formulation involves two key steps: nanoemulsion preparation followed by gel integration. First, a stable nanoemulsion is developed using either high-energy or low-energy emulsification techniques, which is then blended with a gelling agent to form the final nanoemulgel [61].

Step 1: Nanoemulsion Formation

Nanoemulsions can be created using:

- High-energy methods like ultrasonication, high-pressure homogenization, and rotor-stator homogenization, which use mechanical force to reduce droplet size [typically 5–500 nm] and stabilize the system [62].
- Low-energy methods such as spontaneous emulsification, phase inversion composition [PIC], and phase inversion temperature [PIT], which rely on chemical energy and interfacial turbulence to generate nano-sized droplets with minimal external energy.

Ultrasonication

Ultrasound at ~20 kHz breaks droplets into nano-size by applying intense energy. A water jacket maintains temperature, and high shear mixing ensures uniform droplet size distribution.

High-Pressure Homogenization

Involves forcing the formulation through narrow gaps under high pressure [500–5000 psi], using shear, cavitation, and turbulence to produce uniform nanoemulsions. It is scalable and efficient but may cause heat-related degradation [Garg et al., 2016].

High-Speed Homogenization

Rotor-stator homogenizers are widely used in industries for emulsification and particle size reduction. They are cost-effective and suitable for scaling up.

Low-Energy Techniques

Use the system's inherent chemical energy with minimal agitation. These include:

- Spontaneous emulsification: Rapid diffusion of surfactants and co-surfactants between organic and aqueous phases creates turbulence and reduces droplet size.
- Phase Inversion Composition [PIC]: Gradual water addition to oil-surfactant mixtures causes a switch from w/o to o/w emulsions at room temperature using magnetic stirring.
- Phase Inversion Temperature [PIT]: Temperature changes modify surfactant curvature [especially nonionic ones], leading to phase inversion and nanoemulsion formation.



**Kousalya et al.,****Step 2: Nanoemulgel Formation**

The prepared nanoemulsion is incorporated into a gel base by dissolving a polymer [e.g., Carbopol or HPMC] in purified water under mechanical stirring [Hamsinah et al., 2021]. This results in a semi-solid nanoemulgel, combining the stability of a gel with the enhanced delivery of a nanoemulsion, suitable for transdermal application.

Characterization studies of nano-emulgel

To ensure the quality and consistency of nanoemulgel formulations, characterization studies are essential. These evaluations provide insights into the product's stability, efficacy, and behavior. Several universal tests, as outlined by the United States Pharmacopeia [USP], are applicable to any dosage form, including description, identification, assay, and impurity analysis. Additionally, specific tests for topical dosage forms, including nanoemulgels, must be conducted.

Key Parameters in Nanoemulgel Characterization**pH Assessment**

The pH of a nanoemulgel is critical as it affects both skin compatibility and drug release. The pH of human skin generally ranges from 4.5 to 6, and maintaining this range ensures that the formulation is safe and effective for topical use.

Spreadability Evaluation

This parameter refers to the ease with which the gel spreads over the skin. It is assessed by evaluating the "Slip" and "Drag" characteristics, which are indicative of the formulation's application properties.

Droplet Size Analysis

Droplet size is crucial for the stability and effectiveness of nanoemulgels. Dynamic light scattering [DLS] is often used to measure the droplet size and distribution.

Polydispersity Index [PDI]

PDI provides information about the uniformity of droplet size in the formulation. A lower PDI indicates a narrower size distribution, which is essential for consistent performance.

Zeta Potential

Zeta potential helps assess the surface charge of the nanoemulgel, indicating its stability. A high zeta potential indicates good stability, as it prevents aggregation of droplets.

Drug Content

The drug content test determines the total amount of drug in the formulation. This is an essential parameter as higher drug content correlates with minimal loss during manufacturing.

Accelerated Stability Testing

To assess the long-term stability of nanoemulgels, accelerated stability tests are conducted in compliance with ICH guidelines. These studies typically involve storing formulations at temperatures such as $37^{\circ}\text{C} \pm 2^{\circ}\text{C}$, $45^{\circ}\text{C} \pm 2^{\circ}\text{C}$, and $60^{\circ}\text{C} \pm 2^{\circ}\text{C}$ for three months, with periodic analysis of drug content and other quality parameters.

Rheological Characteristics

Rheological studies measure the flow and deformation behaviors of the nanoemulgel when subjected to external forces. This is essential for evaluating the spreadability, drug release, and skin permeation. Rheological properties are evaluated using instruments like Brookfield viscometers and rheometers. The shear-thinning behavior of a formulation, for instance, improves its spreadability and skin permeation.



**Kousalya et al.,**

Applications of nanoemulgels: Curcumin Nanoemulgel: W. Soliman et al. [2021] developed a curcumin-loaded nanoemulgel to improve the bioavailability and skin penetration of curcumin, a poorly water-soluble drug. The formulation exhibited enhanced skin penetration, superior anti-inflammatory effects, and steady transdermal drug release, outperforming traditional curcumin gels.

Tocotrienol-Rich Naringenin Nanoemulgel: E. Yeo et al. [2021] formulated a nanoemulgel to treat chronic wounds in diabetic patients. The formulation showed sustained release over 24 hours, offering more controlled drug release compared to nanoemulsions without polymer coverage. **Finasteride Nanoemulgel for Male Pattern Baldness:** D. Upadhyay et al. [2020] developed a finasteride-loaded nanoemulgel to improve drug permeability through the skin. The formulation led to significant hair growth and enhanced hair thickness compared to the testosterone-treated group.^[64] for Patents on Nano-emulgel Formulations go through table 4.

CONCLUSION

Nanoemulsions represent a non-equilibrium, optically clear, thermodynamically stable, and metastable dispersion of nano-sized particles characterized by a defined surface tension, which is generated through specific shear forces. These dispersions consist of a suitable oil combined with a precise blend of surfactants and co-surfactants, enabling them to solubilize significant amounts of hydrophobic pharmaceuticals. The formation of nanoemulsions can be achieved through various techniques, including the use of homogenizers, low-energy emulsification, and phase inversion methods. Furthermore, there exists considerable debate regarding the optimal approach for preparing nanoemulsions, although it has been demonstrated that they can be effectively formulated using a low-energy emulsification process in conjunction with high shear homogenization. Additionally, nanoemulgel, also referred to as hydrogel-thickened nanoemulsion [HTN], exhibits an increase in viscosity relative to standard nanoemulsions. The stability of a nanoemulsion formulation is further improved by the incorporation of nanoemulgel. It reduces surface and interfacial tension, which in turn leads to an increase in the viscosity of the water-based phase.

Conflict of Interest

The authors declare that there is no conflict of interest regarding the publication of this paper.

ACKNOWLEDGMENTS

The authors have no acknowledgments to declare

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Kousalya et al.,

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Table 1 The Drug Approved By FDA For TDDS.

Sno	Approved Year	Drug Products	Indication
1	1999	Lidocaine	Post hepatic pain
2	2001	Ethinyl estradiol	Contraceptive
3	2003	Oxybutynin	Overactive bladder
4	2006	Fentanyl	Acute postoperative pain
5	2007	Rotigotine	Parkinson's disease
6	2008	Granisetron	Chemo induced emesis
7	2010	Buprenorphine	Chronic pain
8	2022	Donepezil	Alzheimer's disease

Table-2: Enlisted recent literature on nano-carriers-based TDDS research and their outcomes.

Brand Name (Active Ingredient)	Type of Patch	Recommended Duration
Alora (Estradiol)	Matrix-based	Every 3 to 4 days
Androderm (Testosterone)	Membrane-controlled	Once daily (24 hours)
Catapres-TTS (Clonidine)	Membrane-controlled	Once weekly (7 days)
Climara (Estradiol)	Matrix-based	Weekly application
Duragesic (Fentanyl)	Membrane-controlled	Replace every 72 hours
Esclim (Estradiol)	Matrix-based	Apply every 3 to 4 days

Table 3: Literature on Nanoemulsion

Drug	Carrier	Method	Application
Prednicarbat	Phytosphingosine	High Pressure Homogenization	Treatment of atopic dermatitis
Risperidone	PEG400	Nasal delivery for brain targeting	Antipsychotic therapy
Celecoxib	Diethylene glycol	Spontaneous emulsification	Management of Arthritis and osteoarthritis
Praziquantel	Poloxamer	Spontaneous emulsification	Enhanced schistosomicidal activity
Benzathine	Penicillin G Poloxamer	Spontaneous emulsification	Promising system for delivering more soluble drugs



Kousalya *et al.*,

Ampicillin	PEG 400	Solid Dispersion	Protein drug delivery within oil phase
Polyanionic	Poloxamer 188	High Pressure Homogenization	Application in cancer therapy and inflammatory treatment
Primaquine	Poloxamer 188	High Pressure Homogenization	Liquid formulation for targeting latent malaria

Table 4: Recent Patents on Nano-emulgel Formulations.

Patent Number	API	Title	Disease Indication	Inventor	Published year
WO2020240451A1	Brinzolamide	In-situ gelling nano-emulsion of brinzolamide	glaucoma	Hemant Hanumant BHALERAO, Sajeew Chandran	2020
WO2020121329A1	Minoxidil and castor oil	Minoxidil and castor oil nanoemulgel for alopecia	androgenic alopecia	Sudha suresh Dr. Rathodsoniya ramesh devasani	2020
US11185504B2	Aromatase inhibitors	Transdermal non-aqueous nanoemulgels for systemic delivery of aromatase inhibitor	breast cancer	Qatar University	2021
EP3099301B1	Besifloxacin	Besifloxacin for the treatment of resistant acne	Acne vulgaris	Vyome Therapeutics Ltd	2019
WO2020240451A1	Brinzolamide	In-situ gelling nanoemulsion of brinzolamide	glaucoma	Hemant Hanumant BHALERAO, Sajeew Chandran	2020

